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DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING

EEE 306

Power System I Laboratory

Final Project Report

Section: **B1** Group: **06**

Determination Of Bus Fault Current And Breaker Ratings For LLG And LG Faults

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1 ABSTRACT

This project focuses on the comprehensive analysis of bus fault currents and breaker ratings in power systems during Line to Line to Ground (LLG) and Line to Ground (LG) fault scenarios. Using MATLAB, the study involves the development of a detailed power system model and the implementation of fault simulations at various locations within the network.

The primary objectives include the determination of fault currents at critical buses, and selection of appropriate circuit breaker ratings. The outcomes of the analysis provide crucial insights into the optimal selection of circuit breakers capable of interrupting fault currents safely, thus ensuring the reliability and stability of the power system. The project contributes to the broader field of power system engineering by offering a systematic approach to fault analysis and breaker rating determination. The findings not only enhance the understanding of fault behaviors in power systems but also provide practical recommendations for the implementation of protective devices, aligning with industry standards and regulations.

2 INTRODUCTION

Modern power systems demand meticulous planning and analysis to ensure their reliability and resilience against various fault conditions. The determination of fault currents and the selection of appropriate circuit breaker ratings is one among the critical aspects of power system design. This project focuses on addressing these challenges with a specific emphasis on Line to Line to Ground (LLG) and Line to Ground (LG) faults. The analysis involves calculating fault currents at critical buses, considering factors such as fault type, location, and system parameters. Furthermore, the study encompasses the selection of circuit breaker ratings that align with the calculated fault currents, ensuring the effective interruption of fault currents without compromising system stability.

2.1 USE OF SOFTWARE

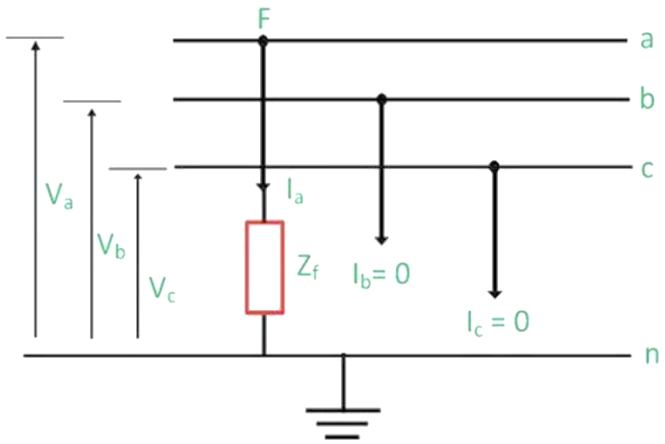
In this project, short circuit fault analysis on power system is done by using MATLAB programming. MATLAB is a powerful software package used for high performance scientific numerical computation, data analysis and visualization.

2.2 NEED FOR FAULT ANALYSIS IN POWER SYSTEM

The system being planned is to be optimal with respect to construction cost, performance and operating efficiency. For this better planning tools are required. In general, the major power system tools are: load Flow Analysis, short circuit analysis or fault calculations, stability analysis etc. The purpose of an electrical power system is to generate and supply electrical energy to consumers with reliability and economy. The greatest threat to this purpose of a power system is the short circuit. The evaluation of fault currents on a power system is significant because the protective devices to be installed on the system depend on the values of the fault currents. Information gained from fault analysis used to select the appropriate size and type of protective equipment to be installed on the system so that the continuity of supply is ensured even when there is a fault on the power system.

2.3 LINE TO GROUND (LG OR SLG) FAULT

In a Single Line to Ground (SLG) fault, a single-phase conductor comes into contact with the ground. This type of fault is characterized by the flow of current from one phase to the ground. SLG faults can occur due to insulation failures, equipment malfunctions, or accidental grounding of a phase conductor. These faults can lead to asymmetrical current flow in the system and pose challenges in terms of fault detection and system protection. Analyzing SLG faults is crucial for maintaining the reliability of power distribution systems and ensuring timely fault isolation to prevent further damage.



Single line-to-ground fault

2.4 DOUBLE LINE TO GROUND (LLG) FAULT

A Line to Line to Ground (LLG) fault involves the simultaneous occurrence of a fault

between two phase conductors and the ground. This fault scenario results in a combination of phase-to-phase and phase-to-ground currents. LLG faults can be caused by factors such as conductor insulation breakdown or equipment failures. Analyzing LLG faults is essential for understanding the complex fault currents that can occur in power systems. The assessment of LLG faults is critical for designing protective measures, such as selecting appropriate breaker ratings, to swiftly and effectively isolate the faulted section and maintain the overall stability of the power distribution network.

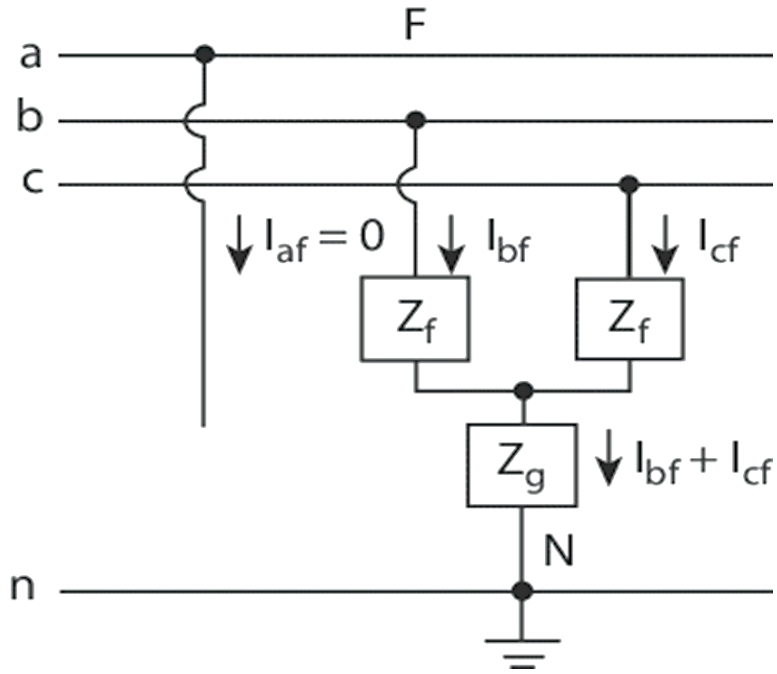


Fig: LLG Fault

3 PROBLEM FORMULATION

The core problem addressed in this project revolves around the determination of bus fault currents and the selection of breaker ratings for LLG and LG faults in power distribution systems. The key aspects to be investigated include:

Fault Current Analysis: Quantifying fault currents at various buses in the distribution network during LLG and LG fault scenarios.

Breaker Sizing: Selecting appropriate breaker ratings to safely interrupt fault currents, considering factors such as interrupting capacity, voltage ratings, and other specifications.

Optimization Recommendations: Providing practical recommendations for optimizing breaker configurations and improving the reliability of the power system.

Addressing these aspects will contribute to the development of effective strategies for managing fault currents and enhancing the overall performance of power distribution networks.

4 FAULT ANALYSIS AND BREAKER RATINGS

4.1 FAULT CURRENT AND ITS CALCULATION

If the insulation of system fails at any point or if two or more conductor that normally operates with a potential difference comes in contact with each other, a short circuit is said to be occur. Fault analysis calculate the fault currents magnitudes that are used to determine short circuit megavolt ampere (SCMVA) ratings of appropriate circuit breakers.

The injected bus currents in terms of bus voltages for a n-bus network is calculated as:

$$\mathbf{I_{bus} = Y_{bus} . V_{bus}} \quad (1)$$

where I_{bus} is the bus current vector entering the bus & Y_{bus} is the bus admittance matrix.

For a fault at bus k, the current entering every bus except the faulted bus k is zero thus equation (1) becomes:

$$\begin{bmatrix} 0 \\ 0 \\ \vdots \\ -\mathrm{Ik(F)} \\ 0 \end{bmatrix} = \begin{bmatrix} Y11 & \dots & Y1k & Y1n \\ Y21 & \dots & Y2k & Y2n \\ \vdots & \ddots & \vdots & \vdots \\ Yk1 & \dots & Ykk & Ykn \\ Yn1 & \dots & Ynk & Ynn \end{bmatrix} \begin{bmatrix} \Delta V1 \\ \Delta V2 \\ \vdots \\ \Delta Vk \\ \Delta Vn \end{bmatrix} \quad (2)$$

Equation (2) can be written as

$$I_{bus}(F) = Y_{bus} . \Delta V_{bus} \quad (3)$$

From the above equation,

$$\Delta V_{bus} = \text{inverse}(Y_{bus}). I_{bus}(F) = Z_{bus} . I_{bus}(F) \quad (4)$$

Where $Z_{bus} = \text{inverse}(Y_{bus})$ is the bus impedance matrix

$$V_{bus}(F) = V_{bus}(0) + \Delta V_{bus} \quad (5)$$

Substituting (4) into equation (5) gives;

$$V(F)= V_{bus}(0)+Z_{bus}.I_{bus}(F) \quad (6)$$

Above equation (6) can be written in matrix form as:

$$\begin{bmatrix} V1(F) \\ V2(F) \\ \vdots \\ Vk(F) \\ Vn(F) \end{bmatrix} = \begin{bmatrix} V1(0) \\ V2(0) \\ \vdots \\ Vk(0) \\ Vn(0) \end{bmatrix} + \begin{bmatrix} Z11 & \dots & Z1k & Z1n \\ Z21 & \dots & Z2k & Z2n \\ \vdots & \ddots & \vdots & \vdots \\ Zk1 & \dots & Zkk & Zkn \\ Zn1 & \dots & Znk & Znn \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ \vdots \\ -\mathrm{Ik(F)} \\ 0 \end{bmatrix}$$

The voltage at bus k during the fault is $V_k(F)$. So;

$$V_k(f) = V_k(0) – Z_{kk}. I_k(F) \quad (7)$$

But $V_k(F)= Z^f.I_k(F)$, where Z^f is fault impedance and Z_{kk} is the kth element of bus impedance matrix and this equation when put in above equation (7), it becomes:

$$Z^f.I_k(F) = V_k(0) – Z_{kk}. I_k(F)$$

Solving for $I_k(F)$ gives:

$$\mathbf{I_k(F) = \frac{Vk(0)}{(Zkk + Zf)}} \quad (8)$$

For bolted or solid fault , $Z^f = 0$. For any bus i the bus voltage during fault is:

$V_i(F) = V_i(0) - Z_{ik} \cdot I_k(F)$, where Z_{ik} is ith kth element of bus impedance matrix.

From equation(8) :

$$\mathbf{V_i(F) = V_i(0) - Z_{ik} \cdot \frac{Vk(0)}{(Zkk + Zf)}} \tag{9}$$

The short circuit current in the line connected between bus i and j is:

$$\mathbf{I_{ij}(F) = \frac{Vi(F) - Vj(F)}{zij}} \tag{10}$$

where z_{ij} is series impedance when line connected between buses i and j.

4.2 SELECTION OF CIRCUIT BREAKERS

The circuit breaker has to perform the following major duties under short circuit conditions:

- To open the contacts to clear the fault.
- To close the contacts onto a fault.
- To carry fault current for a short time while another circuit breaker is clearing the fault.

In addition to the rated voltage, current and frequency, circuit breakers have the following important ratings: 1) Breaking capacity 2) Making capacity 3) Short time capacity

The Breaking capacity of a circuit breaker is of two types i.e. Symmetrical Breaking capacity and Asymmetrical Breaking capacity.

Symmetrical breaking capacity: It is the r.m.s value of the ac component of the fault current that the circuit breaker is capable of breaking under specified conditions of recovery voltage.

Asymmetrical breaking capacity: It is the r.m.s value of the total current comprising of both ac and dc components of the fault current that the CB can break under specified conditions of recovery voltage. The breaking capacity of a CB is generally expressed in MVA. If voltage & current in p.u. values on a 3ph. Basis, then

$$\textbf{SCMVA(3 ph)} = |V_{prefault}| * |I_{sc}| * \text{(MVA base)}$$

Rated MVA interrupting capacity of a CB is to be more than or equal to the short circuit MVA required to be interrupted. The rated asymmetrical breaking capacity is 1.6 times the rated symmetrical current.

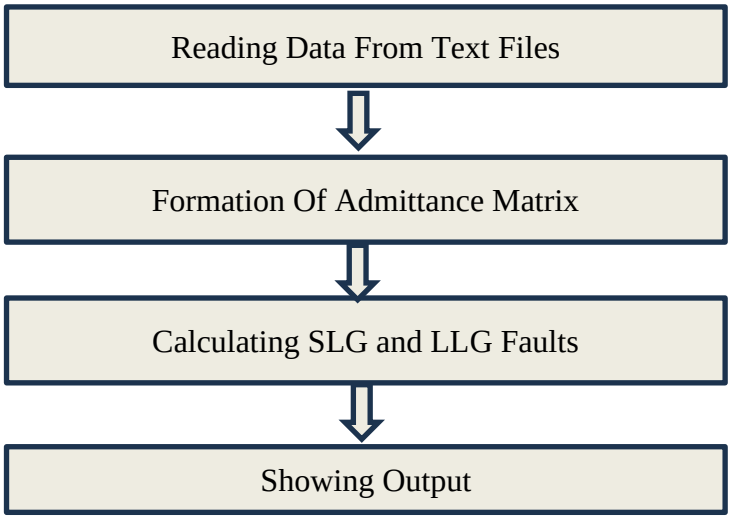
5 METHODOLOGY

5.1 FAULT CURRENT DETERMINATION

This analysis is mainly done in two parts: 1) Data Entry 2) Fault Analysis

Data Entry: Inputs for bus, generators, transformers, and lines are given. Those remain stored in a text file. While running the main program, text files are read using 'textread' function.

Fault Analysis:



For LG Fault:

$$I_{fa}^{(0)} = I_{fa}^{(1)} = I_{fa}^{(2)} = \frac{V_f}{Z_{kk}^{(0)} + Z_{kk}^{(1)} + Z_{kk}^{(2)} + 3Z_f}$$

For LLG Fault:

$$I_{fa}^{(1)} = \frac{V_f}{Z_{kk}^{(1)} + \left[\frac{Z_{kk}^{(2)}\left(Z_{kk}^{(0)} + 3Z_f\right)}{Z_{kk}^{(2)} + Z_{kk}^{(0)} + 3Z_f}\right]}$$

$$I_{fa}^{(0)} = -I_{fa}^{(1)}\left[\frac{Z_{kk}^{(2)}}{Z_{kk}^{(2)} + Z_{kk}^{(0)} + 3Z_f}\right]$$

$$I_{fa}^{(2)} = -I_{fa}^{(1)}\left[\frac{Z_{kk}^{(0)} + 3Z_f}{Z_{kk}^{(2)} + Z_{kk}^{(0)} + 3Z_f}\right]$$

Here, the superscripts 0,1,2 respectively denotes zero, positive, and negative sequences.

6 RESULTS

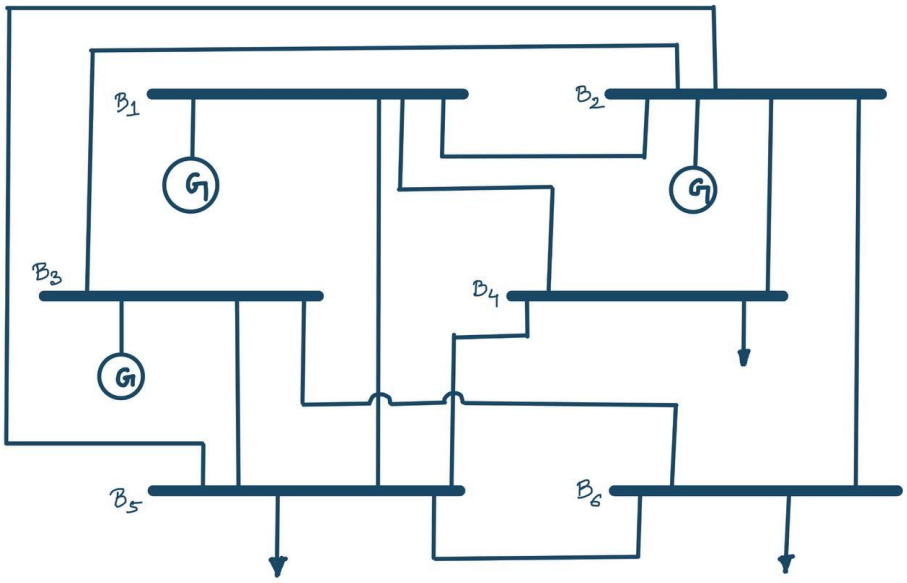
6.1 IEEE-6 BUS SYSTEM

6.1.1 Data: It consists of six bus, three generators, and eleven lines. Details are as follows:

Bus No.	Bus Type	Voltage Profile	P_{gen} (P.U MW)	P_{load} (P.U MW)	Q_{load} (P.U MVAR)
1	Swing	1.05			
2	Gen	1.05	0.50	0.0	0.0
3	Gen	1.07	0.60	0.0	0.0
4	Load		0.0	0.7	0.7
5	Load		0.0	0.7	0.7
6	Load		0.0	0.7	0.7

From Bus	To Bus	R(pu)	X(pu)	BCAP(pu)
1	2	0.10	0.20	0.02
1	4	0.05	0.20	0.02
1	5	0.08	0.30	0.03
2	3	0.05	0.25	0.03
2	4	0.05	0.10	0.01
2	5	0.10	0.30	0.02
2	6	0.07	0.20	0.025
3	5	0.12	0.30	0.025
3	6	0.02	0.10	0.01
4	5	0.20	0.40	0.04
5	6	0.10	0.30	0.03

6.1.2 Single Line Diagram



6.1.3 Code Outputs

LLG Fault Output:

```
Fault current of LLG fault

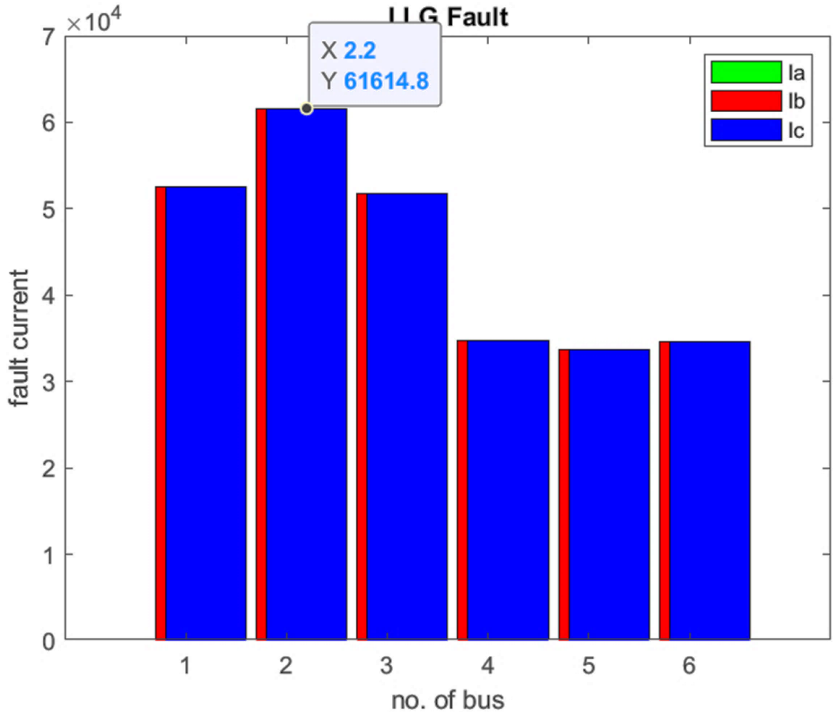
Magnitude(A)    Angle(degree)
Ia
  0.000000    0.000000

Ib
52517.095942 -180.000000

Ic
52517.095942   -0.000000

In
  0.000000 -90.000000
```

LLG Fault Currents for Faults At Different Bus



Thus LLG fault current for fault at bus 1 is 52517 Ampere.

SLG Fault Output:

```
Fault current of SLG fault

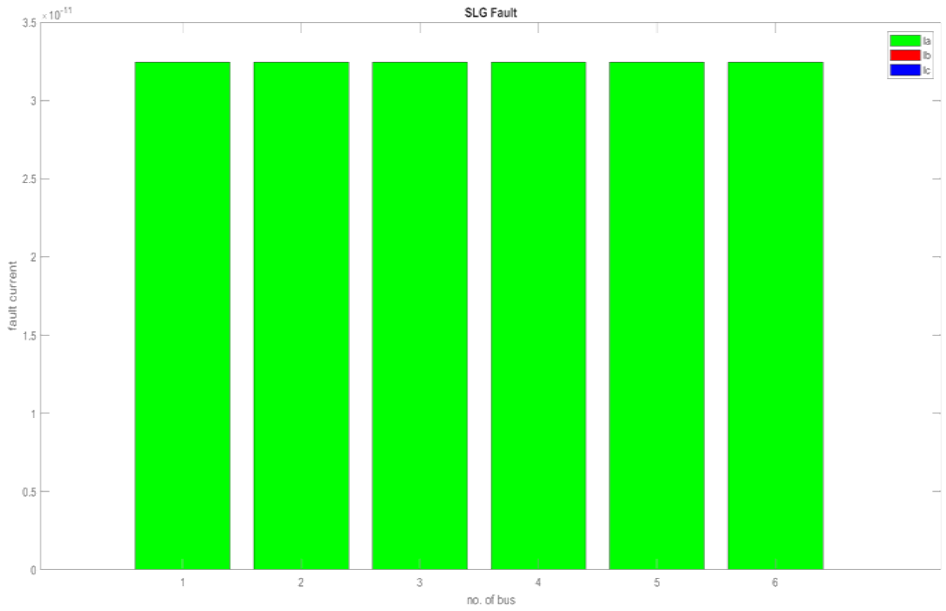
Magnitude(A)    Angle(degree)
Ia
  0.000000   90.000000

Ib
  0.000000    0.000000

Ic
  0.000000    0.000000

In
  0.000000   90.000000
```

SLG Fault Currents for Faults At Different Bus



Thus SLG fault current for fault at all bus is near to zero, in the range of 10^{-11}. It proves that the **severity of LG Fault is comparatively less than LLG Fault**.

Breaker Rating Output:

```
Breaker_rating =

   25.0146

fx >>
```

6.1.4 PSAF Comparison

ID	Prefault kV	Angle	Fault type	Fault S [MVA]	Ia [A]	Ia [deg]	Ib [A]	Ib [deg]	Ic [A]	Ic [deg]	In [A]	In [deg]
Faulted Bus ->												
B1	13.80	0.00	LG	0.0	0.0126	0.0000	0.0000	0.0000	0.0000	0.0000	0.0126	0.0000
B1	13.80	0.00	LLG	1250	0.0014	180.0000	52285.7630	-178.9238	52285.7643	1.0762	0.0000	0.0000

Faulted Bus ->												
B2	13.80	0.00	LG	0.0	0.0126	0.0000	0.0000	0.0000	0.0000	0.0000	0.0126	0.0000
B2	13.80	0.00	LLG	1469	0.0017	180.0000	61477.8920	-179.4464	61477.8937	0.5536	0.0000	0.0000

B3	13.80	0.00	LG	0.0	0.0126	0.0000	0.0000	0.0000	0.0000	0.0000	0.0126	0.0000
B3	13.80	0.00	LLG	1237	0.0014	180.0000	51748.7397	-179.8779	51748.7411	0.1221	0.0000	0.0000

Faulted Bus ->												
B4	13.80	0.00	LG	0.0	0.0126	0.0000	0.0000	0.0000	0.0000	0.0000	0.0126	0.0000
B4	13.80	0.00	LLG	817	0.0041	105.7086	34186.2810	-176.1313	34186.2818	3.8687	0.0000	0.0000

Faulted Bus ->												
B5	13.80	0.00	LG	0.0	0.0126	0.0000	0.0000	0.0000	0.0000	0.0000	0.0126	0.0000
B5	13.80	0.00	LLG	805	0.0009	180.0000	33691.0797	-179.8502	33691.0806	0.1498	0.0000	0.0000

Maximum LLG Fault Current at Bus 1 From Project: 52517 Ampere

Maximum LLG Fault Current at Bus 1 From PSAF: 52285 Ampere

Thus, it is observed that the output of our project matches closely with PSAF output.

6.2 IEEE-14 BUS SYSTEM

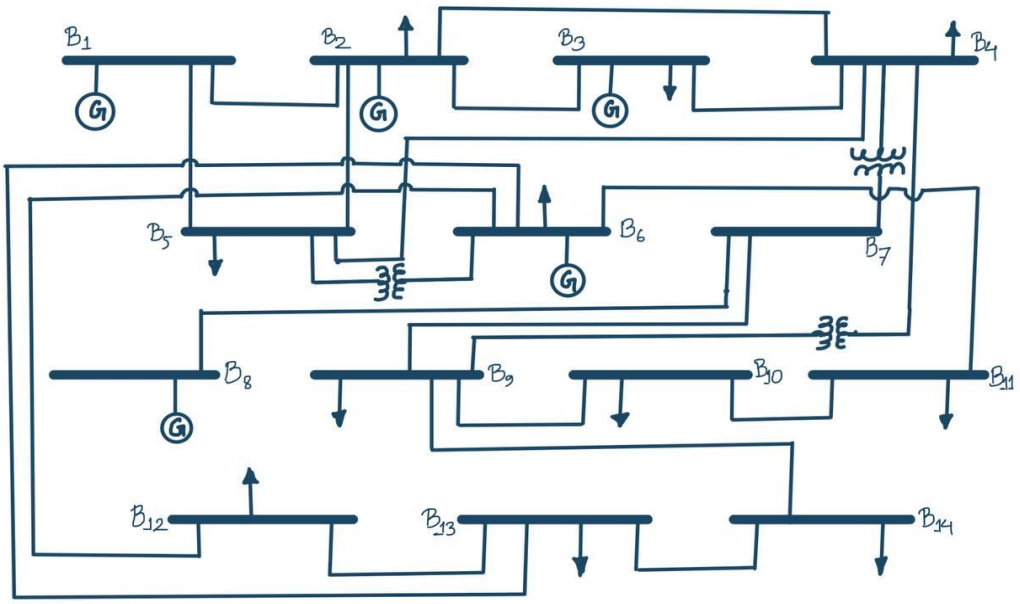
6.2.1 Data: It consists of fourteen bus, five generators, three transformers, and twenty lines. Their impedances are given in the data below:

Bus No.	Bus Voltage		Generation		Load	
	Magnitude Per Unit	Phase Angle Degrees	Real MW	Reactive MVAR	Real MW	Reactive MVAR
1	1.060	0.0	232.4	- 16.9	0.0	0.0
2	1.045	- 4.98	40.0	42.4	21.7	12.7
3	1.010	- 12.72	0.0	23.4	94.2	19.0
4	1.019	- 10.33	0.0	0.0	47.8	3.9
5	1.020	- 8.78	0.0	0.0	7.6	1.6
6	1.070	- 14.22	0.0	12.2	11.2	7.5
7	1.062	- 13.37	0.0	0.0	0.0	0.0
8	1.090	- 13.36	0.0	17.4	0.0	0.0
9	1.056	- 14.94	0.0	0.0	29.5	16.6
10	1.051	- 15.10	0.0	0.0	9.0	5.8
11	1.057	- 14.79	0.0	0.0	3.5	1.8
12	1.055	- 18.07	0.0	0.0	6.1	1.6
13	1.050	- 15.16	0.0	0.0	13.5	5.8
14	1.036	- 16.04	0.0	0.0	14.9	5.0

Transformer	Between Buses	Tap Setting
1	4 - 7	0.978
2	4 - 9	0.969
3	5 - 6	0.932

Line No	Between Buses	Line impedance		Half Line Charging Susceptance per unit
		R per unit	X per unit	
1	1 – 2	0.01938	0.05917	0.02640
2	2 – 3	0.04699	0.19797	0.02190
3	2 – 4	0.05811	0.17632	0.01870
4	1 – 5	0.05403	0.22304	0.02460
5	2 – 5	0.05695	0.17388	0.01700
6	3 – 4	0.06701	0.17103	0.01730
7	4 – 5	0.01335	0.04211	0.0064
8	5 – 6	0.0	0.25202	0.0
9	4 – 7	0.0	0.20912	0.0
10	7 – 8	0.0	0.17615	0.0
11	4 – 9	0.0	0.55618	0.0
12	7 – 9	0.0	0.11001	0.0
13	9 – 10	0.03181	0.08450	0.0
14	6 – 11	0.09498	0.19890	0.0
15	6 – 12	0.12291	0.25581	0.0
16	6 – 13	0.06615	0.13027	0.0
17	9 – 14	0.12711	0.27038	0.0
18	10 – 11	0.8205	0.19207	0.0
19	12 – 13	0.22092	0.19988	0.0
20	13 – 14	0.17093	0.34802	0.0

6.2.2 Single Line Diagram



6.2.3 Code Outputs

LLG Fault Output:

```
Magnitude(KA)    Angle(degree)
Ia
      0.00        0.00

Ib
     64.62     -180.00

Ic
     64.62       -0.00

In
      0.00      -90.00
```

Thus LLG Fault Current for fault at Bus 1 is 64.62 kiloAmpere.

SLG Fault Output:

```
Magnitude(KA)    Angle(degree)
Ia
      0.00       90.00

Ib
      0.00        0.00

Ic
      0.00        0.00

In
      0.00 The fault currents have been showed in the output file
```

Thus SLG fault current for fault at all bus is near to zero. It again proves that the **severity of LG Fault is comparatively less than LLG Fault**.

Breaker Rating Output:

```
Breaker_rating =

   30.7794

fx >>
```

6.2.4 PSAF Comparison

	ID	Prefault kV	Angle	Fault type	Fault S [MVA]	Ia [KA]	Ia [deg]	Ib [KA]	Ib [deg]	Ic [KA]	Ic [deg]	In [KA]	In [deg]
1	Faulted Bus ->												
2													
3	B1	13.80	0.00	LG	0.0	0.0002	0.0000	0.0000	0.0000	0.0000	0.0000	0.0002	0.0000
4	B1	13.80	0.00	LLG	1544	0.0001	-44.9897	64.6156	180.0000	64.6157	0.0000	0.0001	0.0000
5													
77	Faulted Bus ->												
78													
79	B2	13.80	0.00	LG	0.0	0.0002	0.0000	0.0000	0.0000	0.0000	0.0000	0.0002	0.0000
80	B2	13.80	0.00	LLG	1808	0.0001	-45.2265	75.6530	180.0000	75.6531	0.0000	0.0001	0.0000
81													
95	Faulted Bus ->												
96													
97	B3	13.80	0.00	LG	0.0	0.0002	0.0000	0.0000	0.0000	0.0000	0.0000	0.0002	0.0000
98	B3	13.80	0.00	LLG	1306	0.0000	-43.1409	54.6305	180.0000	54.6306	0.0000	0.0001	0.0000
99													
109	Faulted Bus ->												
110													
111	B4	13.80	0.00	LG	0.0	0.0002	0.0000	0.0000	0.0000	0.0000	0.0000	0.0002	0.0000
112	B4	13.80	0.00	LLG	1290	0.0001	-46.3113	53.9598	180.0000	53.9599	0.0000	0.0001	0.0000
113													
126													
127	Faulted Bus ->												
128													
129	B5	13.80	0.00	LG	0.0	0.0002	0.0000	0.0000	0.0000	0.0000	0.0000	0.0002	0.0000
130	B5	13.80	0.00	LLG	1261	0.0001	-46.2855	52.7376	180.0000	52.7377	0.0000	0.0001	0.0000
131													
143	Faulted Bus ->												
144													
145	B6	13.80	0.00	LG	0.0	0.0002	0.0000	0.0000	0.0000	0.0000	0.0000	0.0002	0.0000
146	B6	13.80	0.00	LLG	1114	0.0000	-42.9713	46.5907	180.0000	46.5908	0.0000	0.0001	0.0000
147													
161	Faulted Bus ->												
162													
163	B7	13.80	0.00	LG	0.0	0.0002	0.0000	0.0000	0.0000	0.0000	0.0000	0.0002	0.0000
164	B7	13.80	0.00	LLG	770	0.0001	-45.8546	32.2304	180.0000	32.2305	0.0000	0.0001	0.0000
165													
175	Faulted Bus ->												
176													
177	B8	13.80	0.00	LG	0.0	0.0002	0.0000	0.0000	0.0000	0.0000	0.0000	0.0002	0.0000
178	B8	13.80	0.00	LLG	962	0.0000	-42.8386	40.2637	180.0000	40.2638	0.0000	0.0001	0.0000
179													
187	Faulted Bus ->												
188													
189	B9	13.80	0.00	LG	0.0	0.0002	0.0000	0.0000	0.0000	0.0000	0.0000	0.0002	0.0000
190	B9	13.80	0.00	LLG	639	0.0001	-44.9765	26.7416	180.0000	26.7417	0.0000	0.0001	0.0000
15	Faulted Bus ->												
16													
17	B10	13.80	0.00	LG	0.0	0.0002	0.0000	0.0000	0.0000	0.0000	0.0000	0.0002	0.0000
18	B10	13.80	0.00	LLG	501	0.0001	-45.6209	20.9769	179.9999	20.9770	0.0001	0.0001	0.0000
27	Faulted Bus ->												
28													
29	B11	13.80	0.00	LG	0.0	0.0002	0.0000	0.0000	0.0000	0.0000	0.0000	0.0002	0.0000
30	B11	13.80	0.00	LLG	471	0.0000	-41.5615	19.7152	180.0000	19.7152	0.0000	0.0001	0.0000
31													
39	Faulted Bus ->												
40													
41	B12	13.80	0.00	LG	0.0	0.0002	0.0000	0.0000	0.0000	0.0000	0.0000	0.0002	0.0000
42	B12	13.80	0.00	LLG	406	0.0001	-42.3560	16.9926	179.9999	16.9926	0.0001	0.0001	0.0000
43													
51	Faulted Bus ->												
52													
53	B13	13.80	0.00	LG	0.0	0.0002	0.0000	0.0000	0.0000	0.0000	0.0000	0.0002	0.0000
54	B13	13.80	0.00	LLG	553	0.0001	-44.9021	23.1560	180.0000	23.1561	0.0000	0.0001	0.0000
55													
65	Faulted Bus ->												
66													
67	B14	13.80	0.00	LG	0.0	0.0002	0.0000	0.0000	0.0000	0.0000	0.0000	0.0002	0.0000
68	B14	13.80	0.00	LLG	345	0.0001	-41.8815	14.4476	179.9999	14.4477	0.0001	0.0001	0.0000

LLG Fault Current at Bus 1 From Project: 64.62 kiloAmpere

LLG Fault Current at Bus 1 From PSAF: 64.6156 kiloAmpere

Thus, it is observed that the output of our project matches closely with PSAF output.

6.3 IEEE-30 BUS SYSTEM

6.3.1 Data: It consists of thirty bus, six generators, and forty-one lines. Their impedances are given in the data below:

Bus Data:

Bus No.	Bus Voltage		Generation		Load	
	Magnitude (P.U)	Phase angles (Degrees)	Real MW	Reactive MVAR	Real MW	Reactive MVAR
1	1.0500	0.0	138.48	-2.79	0.0	0.0
2	1.0338	-2.7339	57.56	2.47	21.7	12.7
3	1.0313	-4.6815	0.0	0.0	2.4	1.2
4	1.0263	-5.6077	0.0	0.0	7.6	1.6
5	1.0058	-8.9930	24.56	22.57	94.2	19.0
6	1.0208	-6.4547	0.0	0.0	0.0	0.0
7	1.0069	-8.0244	0.0	0.0	22.8	10.9
8	1.0230	-6.4733	35.0	34.84	30.0	30.0
9	1.0332	-8.0300	0.0	0.0	0.0	0.0
10	1.0183	-9.9268	0.0	0.0	5.8	2.0
11	1.0913	-6.1345	17.93	30.78	0.0	0.0
12	1.0399	-9.4036	0.0	0.0	11.2	7.5
13	1.0883	-8.2049	16.91	37.83	0.0	0.0
14	1.0236	-10.3086	0.0	0.0	6.2	1.6
15	1.0179	-10.3600	0.0	0.0	8.2	2.5
16	1.0235	-9.90280	0.0	0.0	3.5	1.8
17	1.0144	-10.1356	0.0	0.0	9.0	5.8
18	1.0057	-10.9253	0.0	0.0	3.2	0.9
19	1.0017	-11.0615	0.0	0.0	9.5	3.4
20	1.0051	-10.8310	0.0	0.0	2.2	0.7
21	1.0061	-10.4047	0.0	0.0	17.5	11.2
22	1.0069	-10.3936	0.0	0.0	0.0	0.0
23	1.0053	-10.7221	0.0	0.0	3.2	1.6
24	0.9971	-10.8465	0.0	0.0	8.7	6.7
25	1.0086	-10.9074	0.0	0.0	0.0	0.0
26	0.9908	-11.3345	0.0	0.0	3.5	2.3
27	1.0245	-10.6624	0.0	0.0	0.0	0.0
28	1.0156	-6.86710	0.0	0.0	0.0	0.0
29	1.0047	-11.8893	0.0	0.0	2.4	0.9
30	0.9932	-12.7699	0.0	0.0	10.6	1.9

Line Data

Line No	Between Buses	Line impedance		Half Line Charging Susceptance per unit
		R per unit	X per unit	
1	1 – 2	0.0192	0.0575	0.0528
2	1 – 3	0.0452	0.1652	0.0408
3	2 – 4	0.0570	0.1737	0.0368
4	3 – 4	0.0132	0.0379	0.0084
5	2 – 5	0.0472	0.1983	0.0418
6	2 – 6	0.0581	0.1763	0.0374
7	4 – 6	0.0119	0.0414	0.0090
8	5 – 7	0.0460	0.1160	0.0204
9	6 – 7	0.0267	0.0820	0.0170
10	6 – 8	0.0120	0.0420	0.0090
11	6 – 9	0.0	0.2080	0.0
12	6 – 10	0.0	0.5560	0.0
13	9 – 10	0.0	0.1100	0.0
14	4 – 12	0.0	0.2560	0.0
15	12 – 14	0.1231	0.2559	0.0
16	12 – 15	0.0662	0.1304	0.0
17	12 – 16	0.0945	0.1987	0.0
18	14 – 15	0.2210	0.1997	0.0
19	16 – 17	0.0524	0.1923	0.0
20	15 – 18	0.1073	0.2185	0.0
21	18 – 19	0.0639	0.1292	0.0
22	19 – 20	0.0340	0.0680	0.0
23	10 – 20	0.0936	0.2090	0.0
24	10 – 17	0.0324	0.0845	0.0
25	10 – 21	0.0348	0.0749	0.0
26	10 – 22	0.0727	0.1499	0.0
27	21 – 22	0.0116	0.0236	0.0
28	15 – 23	0.1000	0.2020	0.0
29	22 – 24	0.1150	0.1790	0.0
30	23 – 24	0.1320	0.2700	0.0
31	24 – 25	0.1885	0.3292	0.0
32	25 – 27	0.1093	0.2087	0.0
33	28 – 27	0.0	0.3960	0.0
34	27 – 29	0.2198	0.4153	0.0
35	27 – 30	0.3202	0.6027	0.0
36	29 – 30	0.2399	0.4533	0.0
37	8 – 28	0.0636	0.2000	0.4028
38	6 – 28	0.0169	0.0599	0.0130
39	9 – 11	0.0	0.2080	0.0
40	12 – 13	0.0	0.1400	0.0
41	25 – 26	0.2544	0.3800	0.0

6.3.2 Code Outputs

LLG Fault Output:

```
Fault current of LLG fault

Magnitude(KA)    Angle(degree)
Ia
      0.00        0.00

Ib
     67.85      180.00

Ic
     67.85        0.00

In
      0.00       90.00
```

Thus LLG Fault Current for fault at Bus 1 is 67.85 kiloAmpere.

SLG Fault Output:

```
Fault current of SLG fault

Magnitude(KA)    Angle(degree)
Ia
      0.00      -90.00

Ib
      0.00        0.00

Ic
      0.00        0.00

In
      0.00      -90.00
```

Thus SLG fault current for fault at all bus is near to zero. It again proves that the **severity of LG Fault is comparatively less than LLG Fault**.

Breaker Rating Output**:**

```
Breaker_rating =

   32.3179

fx >>
```

6.3.3 PSAF Comparison

	ID	Prefault kV	Angle	Fault type	Fault S [MVA]	Ia [KA]	Ia [deg]	Ib [KA]	Ib [deg]	Ic [KA]	Ic [deg]	In [KA]	In [deg]
1	Faulted Bus ->												
2													
3	B1	13.80	0.00	LG	0.0	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0001	0.0000
4	B1	13.80	0.00	LLG	1619	0.0000	-36.6217	67.7294	180.0000	67.7294	0.0000	0.0000	0.0000
5													

LLG Fault Current at Bus 1 From Project: 67.85 kiloAmpere

LLG Fault Current at Bus 1 From PSAF: 67.7294 kiloAmpere

Thus, it is observed that the output of our project matches closely with PSAF output.

6.4 INPUT PARAMETERS AND ASSUMPTIONS

Only the complex part of line impedance has been considered because ‘textread’ function cannot read both the real and complex part of a complex number. From theoretical calculations, we know that resistance (real part of impedance) can be neglected while calculating fault. So, we only use the reactance part.

The default value of zero sequence impedance has been used for the sake of comparison with PSAF. Other values of positive impedance are given in IEEE bus and line data.

7 MENTIONABLE PROGRAM OUTCOMES OF THE PROJECT

PO(a): Engineering Knowledge: The theoretical knowledge of symmetrical and unsymmetrical faults was implemented in this project. Moreover, prior acquaintance with the software helped in the advancement the project.

PO(b): Problem Analysis: At first, problem formulation was done, we analyzed the hazardous effects of fault and then decided to solve that problem through our project.

PO(c): Design/development Solution: Taking reference from some research papers and books, we designed a methodology of how to solve the formulated problem. We also chose which software will be convenient for using in our project.

PO(e): Modern tool usage: The use of MATLAB R2022b and PSAF 2.81 in this project helped us to be more acquainted with the use of these modern software. Moreover, we got to know some of their assumptions and limitations as well.

PO(g): Environment and sustainability: Our project aims to solve the faults in power system using breakers of suitable ratings. Thus, it is possible to protect environment from the hazardous effects of fault, like sudden fire, etc.

PO(h) Ethics: While doing this project, we didn't copy from any resource; rather we learnt from different research papers and mentioned them in the reference section.

PO(i): Individual work and team work: We learnt the theory of fault analysis individually. On the other hand, reading research papers, coding, verifying the results, preparing presentation slides, etc. were done as a part of team work.

PO(k): Project management and finance: As it was a software-based project, and we used institutional license for using the software MATLAB, we didn't have any expense as a part of project management.

8 DISCUSSION

The outcomes of this project are expected to contribute valuable insights into the intricacies of fault behaviors in power systems, offering practical recommendations for the enhancement of protective devices. By adhering to industry standards and regulations, the project seeks to provide a robust framework for the optimal design and operation of power systems in the presence of LLG and LG faults. This project will be able to recommend breakers of appropriate ratings only by knowing the faulted bus number.

9 REFERENCES

- Daljeet Kaur, *Short Circuit Fault Analysis of Electrical Power System using MATLAB*, March 2014

- John J. Grainger and William D. Stevenson, Jr. *"Power System Analysis"*, McGraw-Hill, 1994

- Adnan Kakilli, *System analysis with the MVA method for Symmetrical Three-Phase Faults*, 2013.

10 CODE SNIPPETS

```
function x=DATAENTRY(nb,ng,nl,nt)

%% entry of generator data
for i=1:ng
    disp('data entry for generator');
    disp(i);
    gen_bus(i)=input('enter the bus number at which the generator is connected:
');
    pos_gen(i)=input('enter the positive/negative sequence impedance of the
generator: ');
    zero_gen(i)=input('enter the zero sequence impedance of the generator: ');
    gen_ground(i)=input('if the generator is grounded enter the grounding
impedance and if ungrounded enter 1000: ');
end
gen_bus; %CHECK
gen=[gen_bus' pos_gen' zero_gen' gen_ground'];
gen=gen'; %each column is a single gen
fid=fopen('gendata.txt','w');
fprintf(fid,'%6.2f %6.2f %6.2f %6.2f\n',gen);
fclose(fid);

%% entry of line data
for i=1:nl
    disp('data entry for line');
    disp(i);
    start_bus(i)=input('enter the starting bus number: ');
    end_bus(i)=input('enter the ending bus number: ');
    pos_line(i)=input('enter the positive/negative sequence impedance of the
line: ');
    zero_line(i)=input('enter the zero sequence impedance of the line: ');



end


linedata=[start_bus' end_bus' pos_line' zero_line'];
linedata=linedata'; %each column is a single line
fid=fopen('linedata.txt','w');

fprintf(fid,'%6.2f %6.2f %6.2f %6.2f\n',linedata);

fclose(fid);

%% entry of transformer data
if nt~=0
for i=1:nt
    disp('data entry for transformer');
    disp(i)
    hv_bus(i)=input('enter the HV bus of the transformer');
    hv_code(i)=input('enter the connection code [0: Y grounded; 1:Y ungrounded;
2:delta');
    lv_bus(i)=input('enter the LV bus of the transformer');
    lv_code(i)=input('enter the connection code [0: Y grounded; 1: Y ungrounded;
2: delta]');
```

```
function Yz = YZERO2(nb, ng, nl, nt)

Yz = zeros(nb);

%% Incorporation of elements corresponding to line impedances
fid = fopen('linedata.txt', 'r');
bb = textread('linedata.txt');
fclose(fid);

x = bb(:, 1);
y = bb(:, 2);
zp = bb(:, 4);
for i = 1:length(x)
    Yz(x(i), x(i)) = Yz(x(i), x(i)) + 1/zp(i);
    Yz(y(i), y(i)) = Yz(y(i), y(i)) + 1/zp(i);
    Yz(x(i), y(i)) = Yz(x(i), y(i)) - 1/zp(i);
    Yz(y(i), x(i)) = Yz(y(i), x(i)) - 1/zp(i);
end

%% incorporation of elements corresponding to transformers
if nt~=0
fid=fopen('transformerdata.txt','r');
bb=textread('transformerdata.txt');
fclose(fid);
x=bb(:,1);
y=bb(:,3);
hvx=bb(:,2);
lvx=bb(:,4);
z=bb(:,6);
for i=1:length(x)
    if hvx(i)==0 & lvx(i)==0 %represents the case of grounded star at both ends
        Yz(x(i),x(i))=Yz(x(i),x(i))+1/z(i);
        Yz(y(i),y(i))=Yz(y(i),y(i))+1/z(i);
        Yz(x(i),y(i))=Yz(x(i),y(i))+(-1)*1/z(i);
        Yz(y(i),x(i))=Yz(y(i),x(i))+(-1)*1/z(i);
    end

    if hvx(i)==0 & lvx(i)==1 %represents the case of grounded star and ungrounded
star
        Yz(x(i),x(i))=Yz(x(i),x(i))+0;
        Yz(y(i),y(i))=Yz(y(i),y(i))+0;
        Yz(x(i),y(i))=Yz(x(i),y(i))+0;
        Yz(y(i),x(i))=Yz(y(i),x(i))+0;
    end

    if hvx(i)==0 & lvx(i)==2 %represents the case of grounded star and delta
        Yz(x(i),x(i))=Yz(x(i),x(i))+1/z(i);
        Yz(y(i),y(i))=Yz(y(i),y(i))+0;
        Yz(x(i),y(i))=Yz(x(i),y(i))+0;
        Yz(y(i),x(i))=Yz(y(i),x(i))+0;
    end

    if hvx(i)==1 & lvx(i)==0 %represents the case of ungrounded star and grounded
star
        Yz(x(i),x(i))=Yz(x(i),x(i))+0;
        Yz(y(i),y(i))=Yz(y(i),y(i))+0;
```

```
function Yp = YPOSITIVE2(nb, ng, nl, nt)

%% Initialize Yp matrix
Yp = zeros(nb);

%% Incorporation of elements corresponding to line impedances

fid=fopen('linedata.txt','r');
bb=textread('linedata.txt');
fclose(fid);
x=bb(:,1);
y=bb(:,2);
zp=bb(:,3);

for i = 1:nl
    from_bus = x(i);
    to_bus = y(i);

    % Add positive sequence impedance to Yp matrix
    Yp(from_bus, from_bus) = Yp(from_bus, from_bus) + 1/zp(i);
    Yp(to_bus, to_bus) = Yp(to_bus, to_bus) + 1/zp(i);
    Yp(from_bus, to_bus) = Yp(from_bus, to_bus) - 1/zp(i);
    Yp(to_bus, from_bus) = Yp(to_bus, from_bus) - 1/zp(i);
end

%% Incorporation of generator elements

gendata = load('gendata.txt');

for i = 1:ng
    gen_bus = gendata(i, 1);
    pos_gen = gendata(i, 2);
    gnd = gendata(i,4);

% Add generator positive sequence impedance to Yp matrix
    Yp(gen_bus, gen_bus) = Yp(gen_bus, gen_bus) + 1/pos_gen;

end

%% Incorporation of transformer elements

if nt~=0
fid=fopen('transformerdata.txt','r');
bb=textread('transformerdata.txt');
fclose(fid);
x=bb(:,1);
y=bb(:,3);
hvx=bb(:,2);
lvx=bb(:,4);
z_pos = bb(:,5);
for i=1:length(x)
    if hvx(i)==0 & lvx(i)==0 %represents the case of grounded star at both ends
        Yp(x(i),x(i))=Yp(x(i),x(i))+1/z_pos(i);
        Yp(y(i),y(i))=Yp(y(i),y(i))+1/z_pos(i);
        Yp(x(i),y(i))=Yp(x(i),y(i))+(-1)*1/z_pos(i);
        Yp(y(i),x(i))=Yp(y(i),x(i))+(-1)*1/z_pos(i);
    end
```

```
function done=SLGFAULT(nb,ng,nl,nt)
%formation of positive sequence Ybus
Yp=YPOSITIVE2(nb,ng,nl,nt);
%formation of zero sequence Ybus
Yz=YZERO2(nb,ng,nl,nt);
%% formation of positive and negative sequence Zbus
Zp=inv(Yp)
Zn=Zp;
%formation of zero sequence Zbus
check=diag(Yz);
k=find(check==0)
if length(k)==0
    Zz=inv(Yz)
elseif length(k)~=0
    Zz=pinv(Yz)
end
%% preparation of pre-fault voltage vectors
xxx=ones(nb);
vp0=xxx(:,1);
yyy=zeros(nb);
vn0=yyy(:,1);
vz0=vn0;
%calculation of fault current
fid=fopen('faultdata.txt','r');
b=textread('faultdata.txt');
fclose(fid);
bus=b(1);
imp=b(3);
ifp=vp0(bus)/((Zp(bus,bus)+Zn(bus,bus)+Zz(bus,bus)+3*imp)*j);
ifn=ifp;
ifz=ifp;
Ia=ifz+ifp+ifn
Ib=ifz+ifp*((-1/2)+(-sqrt(3)/2)*j)+ifn*((-1/2)+(sqrt(3)/2)*j)
Ic=ifz+ifp*((-1/2)+(sqrt(3)/2)*j)+ifn*((-1/2)+(-sqrt(3)/2)*j)
In=Ia+Ib+Ic
Breaker_rating = 220*sqrt(3)*Ib*10^(-6)*1.25
fprintf('Fault current of SLG fault\n\n');
fprintf('Magnitude(KA)   Angle(degree)\n');
fprintf('Ia\n');
fprintf('%10.2f %10.2f\n\n',abs(Ia)*4.183,angle(Ia)*180/pi);
fprintf('Ib\n');
fprintf('%10.2f %10.2f\n\n',abs(Ib)*4.183,angle(Ib)*180/pi);
fprintf('Ic\n');
fprintf('%10.2f %10.2f\n\n',abs(Ic)*4.183,angle(Ic)*180/pi);
fprintf('In\n');
fprintf('%10.2f %10.2f\n\n',abs(In)*4.183,angle(In)*180/pi);
done=1;
```

```
function done=LLGFAULT(nb,ng,nl,nt)
%formation of positive sequence Ybus
Yp=YPOSITIVE2(nb,ng,nl,nt);
%formation of zero sequence Ybus
Yz=YZERO2(nb,ng,nl,nt);
%formation of positive and negative sequence Zbus
Zp=inv(Yp);
Zn=Zp;

%% formation of zero sequence Zbus
check=diag(Yz);
k=find(check==0)
if length(k)==0
    Zz=inv(Yz)
elseif length(k)~=0
    Zz=pinv(Yz) %singular matrix ke inverse korte use kori
end

%preparation of pre-fault voltage vectors
xxx=ones(nb);
vp0=xxx(:,1); %VOLTAGE LEVELS
yyy=zeros(nb);
vn0=yyy(:,1);
vz0=vn0;

%% calculation of fault current
fid=fopen('faultdata.txt','r');
b=textread('faultdata.txt');
fclose(fid);
bus=b(1);
imp=b(3);

%% calculation of Thevenin impedances
Zth=((Zz(bus,bus)+3*imp)*j*Zn(bus,bus)*j/(((Zz(bus,bus)+3*imp+Zn(bus,bus))*j))
+Zp(bus,bus)*j);
ifp=vp0(bus)/Zth;
ifn=-(ifp*(Zz(bus,bus)+3*imp)*j)/((Zz(bus,bus)+3*imp+Zn(bus,bus))*j);
ifz=-(ifp*(Zn(bus,bus))*j)/((Zz(bus,bus)+3*imp+Zn(bus,bus))*j);
Ia=ifz+ifp+ifn
Ib=ifz+ifp*((-1/2)+(-sqrt(3)/2)*j)+ifn*((-1/2)+(sqrt(3)/2)*j)
Ic=ifz+ifp*((-1/2)+(sqrt(3)/2)*j)+ifn*((-1/2)+(-sqrt(3)/2)*j)
In=Ia+Ib+Ic
fprintf('Fault current of LLG fault\n\n');
fprintf('Magnitude(KA)   Angle(degree)\n');
fprintf('Ia\n');
fprintf('%10.2f %10.2f\n\n',abs(Ia)*4.18369,angle(Ia)*180/pi);
fprintf('Ib\n');
fprintf('%10.2f %10.2f\n\n',abs(Ib)*4.18369,angle(Ib)*180/pi);
fprintf('Ic\n');
fprintf('%10.2f %10.2f\n\n',abs(Ic)*4.18369,angle(Ic)*180/pi);
fprintf('In\n');
fprintf('%10.2f %10.2f\n\n',abs(In)*4.18369,angle(In)*180/pi);

done=1;
```

```
clear all;
clc;
%% main program for the fault analysis package
disp('DETERMINATION OF FAULTS IN POWER SYSTEM');
nb=input('enter the number of buses: ');
ng=input('enter the number of generators: ');
nl=input('enter the number of lines: ');
nt=input('enter the number of transformers: ');

%% the following section determines choice of function
choice=input('enter your choice 1) data entry 2) fault analysis\n');
if choice==1
    DATAENTRY(nb,ng,nl,nt);

end


if choice==2

    %% entry of fault data
    fault_bus=input('enter the faulted bus number: ');

    disp('1)Single line to ground fault');

    disp('2)Line to line fault with ground');

    fault_type=input('Enter the type of fault: ');
    fault_imp=input('Enter the fault impedance: ');
    fault=[fault_bus fault_type fault_imp];
    fid=fopen('faultdata.txt','w');
    fprintf(fid,'%6.2f %6.2f %6.2f\n',fault);
    fclose(fid);

    fid=fopen('faultdata.txt','r');
    b=textread('faultdata.txt');
    fclose(fid);

    if b(2)==1
        %SLG fault analysis
        done=SLGFAULT(nb,ng,nl,nt);
    end

    if b(2)==2
        %LLG fault analysis
        done=LLGFAULT(nb,ng,nl,nt);
    end
end

%%Check if the code ran
disp('The fault currents have been showed in the output file');
```